EVOLUTION OF THE INDO-EUROPEAN LANGUAGE FAMILY: A COMPUTATIONAL SIMULATION

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ABSTRACT

Spoken across much of Eurasia from Western Europe to India and Siberia, Indo-European is the world's largest language family by number of speakers. Its prominence arose through extensive migrations of ancestral populations, but because these migrations occurred before recorded history, they remain only partially understood. We develop a computer program[1] that evaluates theories of the family's origin and expansion by simulating migrations from a given initial homeland. Our results support the Steppe hypothesis, which places the origin of Indo-European in the grassland plains between the Black and Caspian Seas, over the Anatolian hypothesis, which locates it in present-day Turkey, though neither simulation fully recovers the attested relationships among descendant languages.

1 INTRODUCTION

Many of the world's most widely spoken languages—including English, Spanish, German, Russian, Hindi, and Persian—share fundamental similarities in vocabulary and grammar and are therefore believed to stem from a common ancestor. This ancient source language, known as **Proto-Indo-European (PIE)**, has been studied by linguists and historians since the 19th century (Bopp & Windischmann, 1816; Schleicher, 1866; Saussure, 1879), leading to substantial progress in its reconstruction, although some aspects remain uncertain or speculative.

By the time of the earliest written records—the Hittite cuneiform inscriptions (1700 BC), the Mycenaean Greek inscriptions (1400 BC), and the Rigveda (1500–1000 BC)—PIE had already diversified into many mutually unintelligible daughter languages. As a result, there is no direct attestation of what it sounded like or how its evolution into the daughter languages unfolded. Instead, scholars have approached these questions through evidence from comparative linguistics, archaeology, and genetics. Based on the abundance of terms for domesticated animals in the reconstructed lexicon, together with the relative scarcity of terms for crop plants, Proto-Indo-Europeans are generally thought to have been nomadic pastoralists. The domestication of the horse (**h₁éḱwos*) and the use of wheeled wagons (**weǵʰnos*) enabled long-distance mobility and the eventual spread of Indo-European languages throughout Eurasia. The location of the PIE homeland, or *Urheimat*, however, remains debated. The most widely accepted account is the **Steppe hypothesis** (Gimbutas, 1956), which places the *Urheimat* in the Pontic-Caspian steppe of present-day southern Ukraine and southwestern Russia around 4000 BC. A minority of scholars favor the **Anatolian hypothesis** (Renfrew, 1990), according to which Indo-European speakers dispersed out of present-day Turkey starting around 7000 BC. Several other hypotheses have been proposed but are not considered scientifically credible.

We introduce a computational simulation framework that models the spread of PIE speakers from a specified starting location and the evolution of the emerging descendant languages. Our framework implements geography-influenced population diffusion, lexical innovation, and borrowing, and produces a matrix of lexical similarities between the newly formed Indo-European languages. A simulation that starts at a plausible homeland should yield a matrix that aligns closely with attested (actual) similarities. We can thus compare the outcomes of the Steppe and Anatolian models, contributing a bit of evidence to the long-standing *Urheimat* debate. Our Steppe model correctly recovers

[1] Code: https://github.com/dariakryvosheieva/indo-european-migration.

many features of the Indo-European phylogenetic tree, including the early separation of Hittite and Tocharian and the existence of the Balto-Slavic, Indo-Iranian, and Graeco-Albanian clades, although it notably mis-groups Latin and Armenian. The Anatolian model produces an overall-similar grouping, but fails to recover Hittite as a divergent branch due to its location at the origin of the family, confirming a common argument against the Anatolian hypothesis. In summary, our simulations provide additional computational support for the Steppe hypothesis as the more plausible account of Indo-European dispersal.

2 Background

2.1 Indo-European Migrations

2.1.1 Steppe Hypothesis

The identification of PIE speakers with nomads of the Pontic-Caspian steppe was proposed by early scholars (Benfey, 1869; Schrader, 1890) and later systematized by Marija Gimbutas, who associated Proto-Indo-Europeans with the **Yamnaya** archaeological culture (Gimbutas, 1956). This culture extended across much of the steppe region from the Ural mountains to the Danube valley during the fourth and third millennia BC.

According to the Steppe theory, speakers of early Indo-European dialects expanded outward from the Pontic-Caspian steppe in several waves. The first branch, ancestral to Hittite and other closely related languages such as Luwian, moved southwest through the Balkans and eventually into Anatolia after 4000 BC. The second branch traveled eastward, giving rise to the Afanasievo culture and the Tocharian languages circa 3500 BC. A later northwest expansion linked to the Corded Ware culture (3000-2500 BC) resulted in the development of the Balto-Slavic, Germanic, Italic, and Celtic branches. Another movement occurring around 2000 BC founded the Sintashta culture connected with the Proto-Indo-Iranian language, which then split into the Indic and Iranian branches around 1800-1600 BC. Greek, Albanian, and Armenian may share more recent ancestry with one another than with the rest of Indo-European, but this is contested and their exact paths of dispersal are unknown. The hypothesized migrations of steppe populations are summarized in Figure 2.

The Steppe hypothesis enjoys strong archaeological and genetic support. Archaeologically, it is consistent with the existence of steppe-derived cultures corresponding to intermediate migration stages, including the Corded Ware and Sintashta cultures, as well as with similarities between artifacts found in later Indo-European-speaking societies and those produced by Yamnaya people (Anthony, 2007). Genetically, DNA studies reveal substantial steppe ancestry in modern Indo-European-speaking populations in Europe and South Asia, most notably the prevalence of the Y-chromosomal haplogroups R1a-M417 and R1b-M269 of Yamnaya origin (Underhill et al., 2014; Myres et al., 2010).

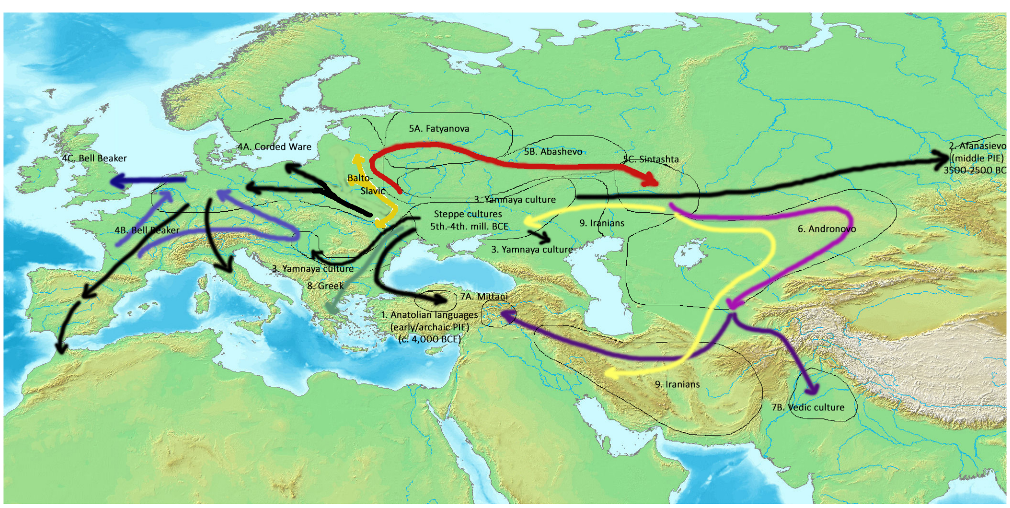

Figure 1: Indo-European migrations according to the Steppe hypothesis.

2.1.2 Anatolian Hypothesis

The main alternative to the Steppe hypothesis is the Anatolian hypothesis, which proposes that Indo-European languages expanded out of Asia Minor along with the spread of farming during the Neolithic agricultural revolution of the 7th and 6th millenia BC (Renfrew, 1990). Under this account, PIE speakers split into a group that stayed in Asia Minor, later developing into the Hittite-Luwian branch, and another one that traveled to Europe. The latter group gave rise to all core Indo-European branches through gradual demic diffusion driven by settled farmers.

Several arguments speak in favor of the Anatolian theory:

- Hittite and Luwian lack certain cognates associated with a mobile lifestyle, which is consistent with Indo-European society being non-nomadic at the time of their separation (Renfrew, 1999);
- Major innovations (such as farming) are likely to cause large-scale demographic turnovers, leading to the replacement or assimilation of most of Europe's indigenous population by Indo-Europeans;
- Agriculture-linked expansions of language families may have been common in the Neolithic world, and have also been proposed for the Afro-Asiatic and Dravidian families (Bellwood & Renfrew, 2002).

However, acceptance of the theory by current scholarship is limited due to counterarguments:

- Agriculture was probably invented in the Levant, with Anatolia serving merely as a corridor for its spread into Europe (Pinhasi et al., 2005);
- The reconstructed PIE vocabulary includes terms related to technologies and practices that postdate the Neolithic revolution, including metallurgy, weaving, and dairy production (Mallory & Adams, 2006);
- The hypothesis needs explicit reconciliation with the Steppe hypothesis (such as assuming adoption of a PIE dialect by nomads or adoption of a nomadic lifestyle by PIE farmers in the steppe) to accomodate substantial evidence in favor of steppe migrations (Piazza & Cavalli-Sforza, 2006).

2.2 Computational Models of Language Change

Modern research increasingly leverages computational models to study how linguistic innovations spread through speaker populations and to evaluate whether proposed mechanisms can reproduce observed patterns of language variation and divergence.

Bel-Enguix (2010) and Buzato & Cunha (2024) both considered graph-based social network models, where nodes represent individual speakers and edges represent social interactions between them. Both works analyzed how language change depends on *prestige*—a parameter that represents a speaker's social status asociated with wealth, power, or authority. Bel-Enguix (2010) divided the population into two groups—one with high prestige and one with low prestige—while Buzato & Cunha (2024) assigned each individual their own prestige value. Language change was modeled as the transmission of innovations from higher-prestige to lower-prestige speakers.

The Belgian computational linguist Peter Dekker conducted a series of case studies on how computational models can explain particular historical instances of language change. In Dekker et al. (2025), he explored the loss of verbal morphology in Alorese, an Austronesian language spoken in Eastern Indonesia. Agents representing adult native speakers dropped a verbal suffix whenever it would cause a consonant cluster, and agents representing language learners generalized the suffix disappearance to words without consonant clusters with a certain probability controlled by the generalization parameter. In Dekker et al. (2024), he studied the effect of conversational priming on language change using pronouns as an example: a question containing a first-person pronoun expects an answer containing a second-person pronoun and vice versa ("Are you there?—Yes, I am there"), while a question containing a third-person pronoun expects an answer containing the same third-person pronoun ("Is he there?—Yes, he is there"). Therefore, speakers should be more likely to remain faithful to their own idiolectal first- and second-person forms, while repeating third-person forms after other speakers. A computational simulation showed that third-person pronouns were the fastest to converge to the same form for all speakers, which could be either the innovative or the conservative form depending which one was designated more prestigious, matching the real-world cross-linguistic tendency of third-person forms to be on the extremes of the change spectrum (either the most innovative or the most conservative).

Baxter et al. (2009) used computational simulation to evaluate the validity of an existing theory, namely Peter Trudgill's theory of the emergence of New Zealand English (Trudgill, 2004). Trudgill divided the evolution of New Zealand English into three stages corresponding to the first three generations of British immigrants into New Zealand: Stage I eliminates the rarest linguistic variants; Stage II eliminates rare linguistic variants up to approximately 10% frequency; Stage III consolidates the majority variant out of those remaining. An attempt to reproduce the theory computationally yielded moderate success, indicating that the theory is plausible but simplistic, and further factors (such as unequal probabilities of accepting linguistic innovations) need to be taken into account to describe the formation of the new dialect more accurately.

3 Methods

Our simulation begins by constructing a stylized geographic grid of Eurasia and seeding a small initial population at a hypothesized homeland location. This population then evolves forward in time for a fixed number of years, allowing speakers to diffuse across neighboring cells, undergo lexical innovation, and borrow lexical items from nearby populations. At the end of the simulation, we sample lexicons from locations corresponding to historically attested Indo-European branches. We compute pairwise lexical similarities between these sampled languages and use the resulting similarity matrix to construct a heatmap and an inferred family tree.

3.1 Geography

We represent Eurasia as a simplified two-dimensional latitude-longitude grid (Figure 2). The map covers latitudes 8°N–72°N and longitudes 15°W–110°E, with each grid cell corresponding to a 1° by 1° square.

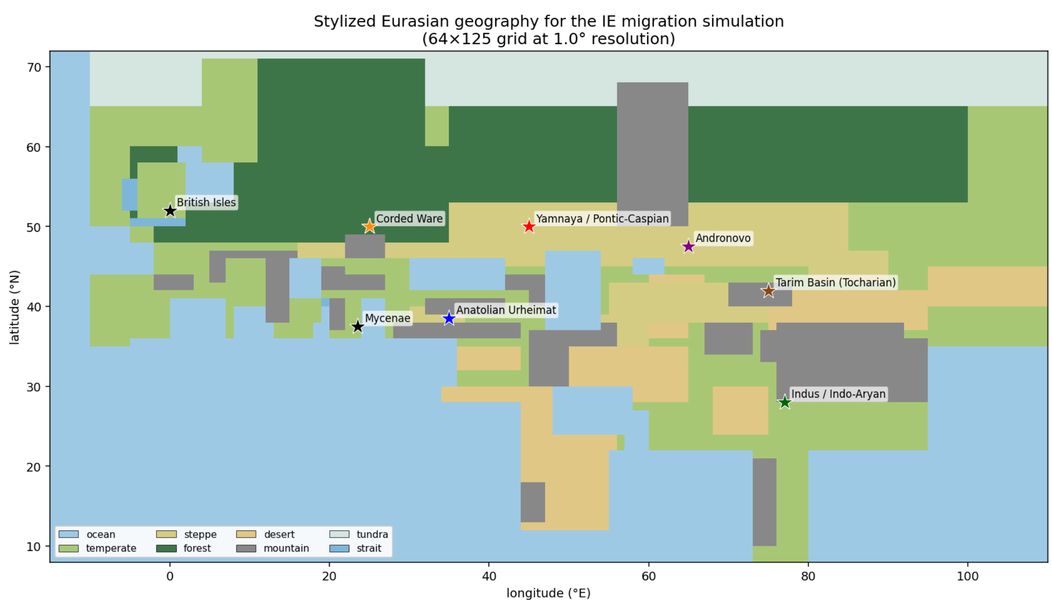

Figure 2: Stylized map of Eurasia for the migration simulation.

Each cell is assigned one of eight terrain types: *steppe*, *forest*, *mountain*, *desert*, *tundra*, *temperate*, *ocean*, or *strait*. Each terrain type has two associated parameters: **capacity**, denoting relative amount of population a cell can support, and **friction**, denoting the difficulty of moving into the cell. For example, steppe cells have low friction and moderate capacity, while mountain and desert cells have higher friction and lower capacity. Ocean cells have zero capacity and infinite friction, while strait cells have a small but nonzero capacity and a large but finite friction (needed for technical reasons). Exact values of capacity and friction for each terrain type are presented in Table 1.

Terrain	Capacity	Friction
Steppe	0.50	1.0
Forest	0.70	2.0
Mountain	0.10	5.0
Desert	0.05	3.0
Tundra	0.10	2.5
Temperate	1.00	1.5
Ocean	0.00	$+\infty$
Strait	0.10	3.0

Table 1: Capacities and frictions of the terrain types used in the map.

The map models the major geographic features relevant to Indo-European migration. The steppes are represented as low-friction corridors stretching from Eastern Europe into Central Asia. Additional passable corridors are included through the Pannonian plain, the Iranian plateau, Bactria-Margiana-Sogdiana, Dzungaria, and the Anatolian plateau. Mountain ranges, including the Pyrenees, Alps, Carpathians, Caucasus, Urals, Zagros, Elburz, Hindu Kush, Pamirs, Tien Shan, Kunlun, and Himalayas, are represented as high-friction cells. The map also features significant water bodies, such as the Mediterranean, Adriatic, Aegean, Black, Caspian, Aral, and Baltic Seas. Broad ecological zones such as forest, tundra, desert, and steppe are placed to approximate Eurasian biomes (Olson et al., 2001).

3.2 Migrations

The initial population of speakers is seeded at 50°N 45°E for the Steppe scenario or 38°N 35°E for the Anatolian scenario. In both cases, the size of the initial population is equal to half of the carrying capacity of its cell. The simulation is then run forward in discrete time steps of length dt, for $T = 4000$ years in the Steppe scenario or $T = 7000$ years in the Anatolian scenario.

Eurasia is treated as a metapopulation: each grid cell has a scalar population count and, once settled, an associated lexicon. At every time step t, population growth, migration, settlement of new cells, and lexical evolution are applied in sequence:

1. **Logistic population growth.** Each cell c grows toward its terrain-specific capacity:

$$N_c^{t+1} = N_c^t \left(1 + r - r\frac{N_c^t}{K}\right) dt,$$

   where N_c^t is the population of cell c in year t, r is the annual growth rate parameter (default 0.01), K is the cell's carrying capacity determined by its terrain type, and dt is the step size (default 10 years).

2. **Migration to eight neighbors.** A fraction m_{eff} of cell c's population is distributed among the eight neighboring cells, including both cardinally and diagonally adjacent cells:

$$m_{\text{eff}} = \min(m \cdot dt, 1.0),$$

   where m is the annual migration rate parameter (default 0.04). The share of migrants sent to each neighbor is inversely proportional to the friction of the destination cell.

3. **Founding new language varieties.** If a cell's population exceeds a fixed fraction (default 5%) of its capacity, it receives a copy of the lexicon of the neighboring source population that contributed the largest inflow during the current time step. From now on, this lexicon will evolve independently. (Migration into an already-founded cell changes its population count but does not replace its lexicon.)

4. **Lexical evolution.** Every founded cell's lexicon mutates via internal innovation and borrowing from nearby cells. Innovation is implemented as replacement of an inherited cognate by a newly-created word form, while borrowing is implemented as contact-induced replacement, in which a cell copies the word form used by a neighboring population for the same meaning slot. See Section 3.3 for further details.

3.3 Lexicon

Every founded cell has its own language variety represented by a lexicon of meaning slots, analogous to a Swadesh-style word list. In our experiments, the number of meaning slots is 1500, approximately matching the number of reconstructed unique PIE roots. Each meaning slot stores an integer cognate-class ID. Two language varieties are treated as sharing a cognate for a given meaning if their integer IDs for that slot are identical; otherwise, their words for that meaning are treated as non-cognate.

For each meaning slot, innovation occurs independently with probability

$$p_i = 1 - \exp(-\lambda \cdot dt),$$

where λ is the annual innovation rate parameter (default $1.5 \cdot 10^{-4}$, chosen so that about 14% of the lexicon is replaced per millennium, matching the glottochronological average). If innovation occurs, the current cognate ID is replaced by a new ID that has never previously appeared in the simulation. This process models ordinary lexical replacement through semantic shift, taboo replacement, or new coinage.

Borrowing is performed after innovation. If neighboring lexicons are available, each non-innovated slot may be replaced by the corresponding word from a neighboring cell. The probability of borrowing is

$$p_b = 1 - \exp\left(-\mu \cdot dt \cdot \sum_k w_k\right),$$

where μ is the annual borrowing rate parameter (default $5 \cdot 10^{-5}$), and $\sum_k w_k$ is the sum, over the eight neighboring cells k, of the borrowing weights w_k of those cells. The borrowing weights depend on population and distance:

$$w_k = N_k^t \exp\left(\frac{-d_k}{\sigma}\right),$$

where d_k is the distance in kilometers to cell k (which depends on whether cell k is cardinally or diagonally adjacent to the current cell, as well as how far away the cells are from the equator), and σ is the contact decay parameter (default 300 km). When borrowing occurs, the source neighbor is sampled in proportion to its w_k.

3.4 Output

At the end of a simulation, every founded cell contains a population count and an associated lexicon. To compare the simulated outcome with the attested Indo-European family, we sample representative lexicons from locations corresponding to major Indo-European languages spoken in the first millennium BC. Each branch is assigned an approximate earliest-attestation location, such as Hittite in central Anatolia, Tocharian in the Tarim Basin, Greek in the Aegean, Latin in central Italy, and Sanskrit in northern India. The sampling locations are plotted on the map in Figure 3.

We then compute a pairwise cognate-distance matrix over the sampled branch lexicons. For two sampled languages A and B, lexical distance is defined as the fraction of meaning slots in which their cognate IDs differ.

Finally, we use the UPGMA methodology (Sokal et al., 1958) to infer the simulated Indo-European family tree from the distance matrix.

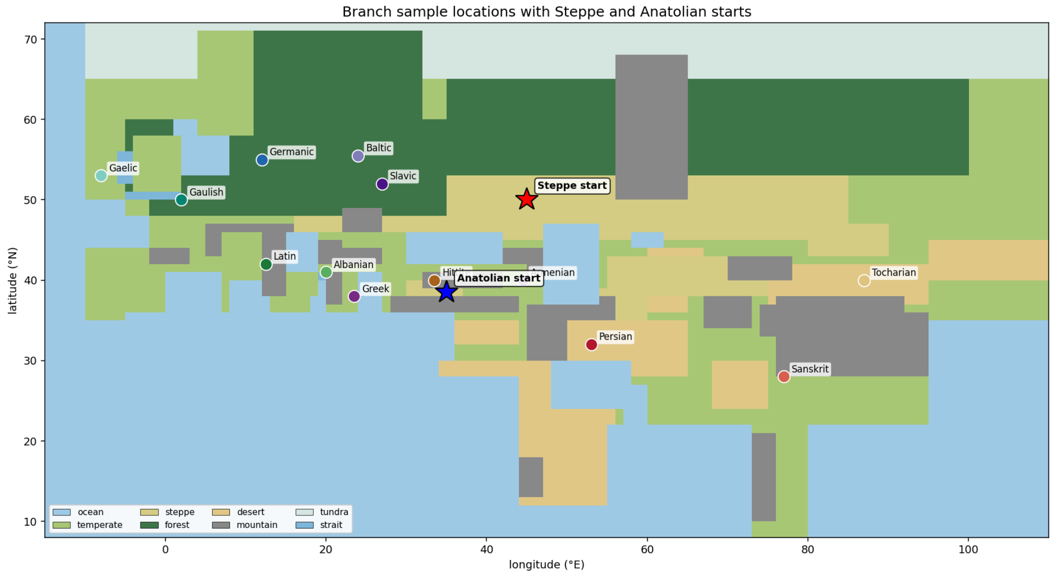

Figure 3: Sampling locations for cells representing historically attested Indo-European languages.

4 Results

4.1 Steppe Model

The results of the Steppe simulation are shown in Figures 4 (lexical distance heatmap) and 5 (family tree). We observe that, due to its geography-based nature, the simulation is biased toward grouping languages together when they are geographically close: for example, the simulation groups Hittite with Persian and Tocharian with Sanskrit, although in reality, Hittite and Tocharian are believed to be the earliest and second-earliest (respectively) languages to diverge from the rest of the family.

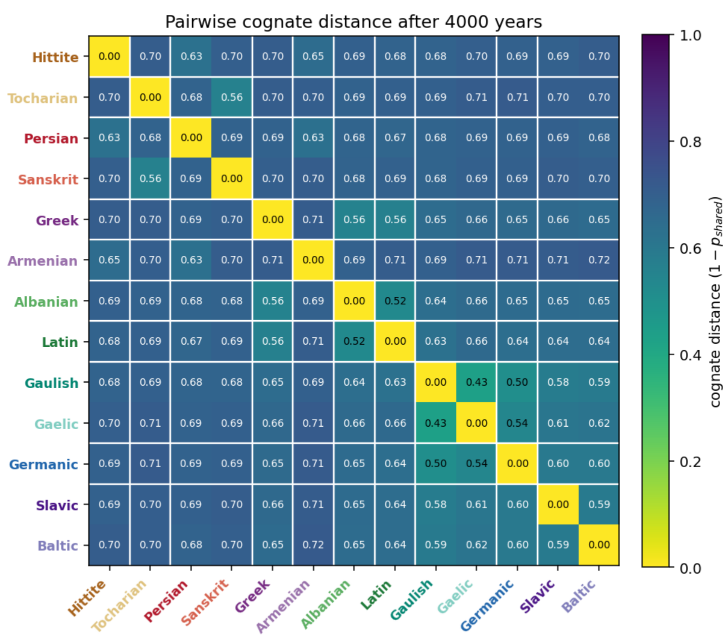

Figure 4: Heatmap of lexical distances at the end of the Steppe simulation.

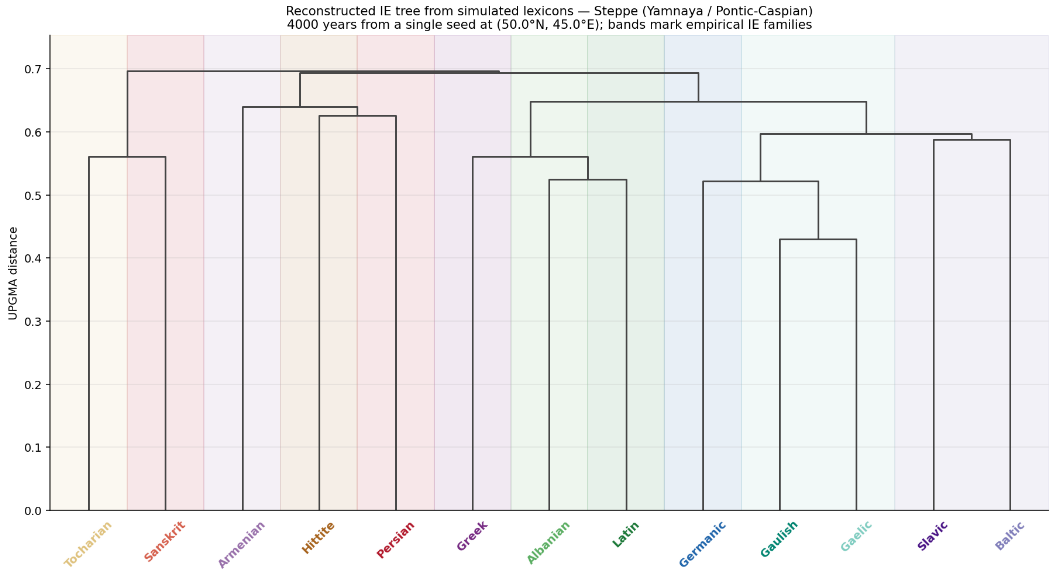

Figure 5: Dendrogram of Indo-European languages inferred from the heatmap in Figure 4.

To solve the issue, we turn to explicit modeling of the early separations of the Hittite and Tocharian branches. At the start of the 4000-year simulation period, instead of initializing the entire PIE population in one cell, we initialize the Hittite and Tocharian populations in their attested locations, along with the core PIE population in the steppe location. The Hittite lexicon is given 1500 years of pre-evolution via innovation, and the Tocharian lexicon is similarly given 500 years of pre-evolution (matching the hypothesized time periods by which these branches predate the main Indo-European expansion). The resulting heatmap and tree diagram are shown in Figures 6 and 7.

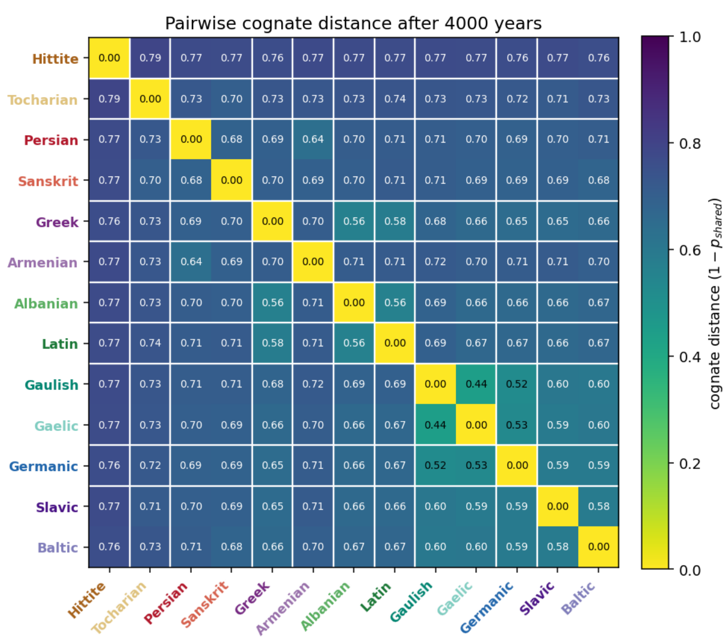

Figure 6: Heatmap of lexical distances at the end of the Steppe simulation if Hittite and Tocharian are pre-evolved.

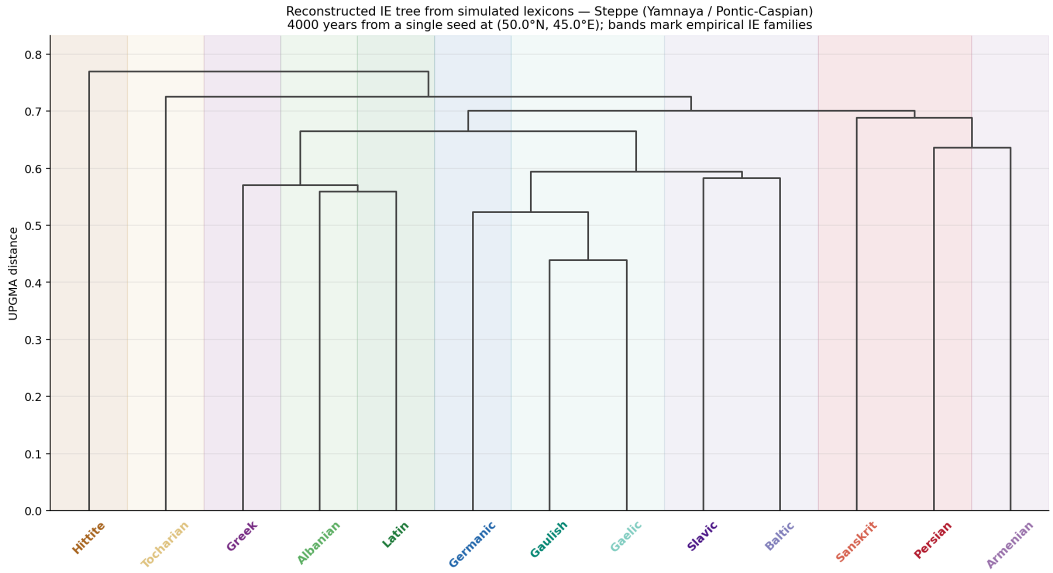

Figure 7: Dendrogram of Indo-European languages inferred from the heatmap in Figure 6.

The revised simulation now *perfectly* recovers a reference phylogenetic tree by Chang et al. (2015) shown in Figure 8, apart from only three discrepancies: (i) Latin is grouped with Greek and Albanian instead of Celtic (Gaulish/Gaelic); (ii) Armenian is grouped with the Indo-Iranian languages instead of Greek and Albanian; (iii) the Graeco-Albanian clade is a sister clade of Corded Ware (Balto-Slavic/Germanic/Celtic) and their union is a sister clade of Indo-Iranian, while in the reference tree, Indo-Iranian is a sister clade of Corded Ware and their union is a sister clade of Graeco-Albanian. We attribute these discrepancies to the (overly) geography-centric nature of the simulation, as it does not account for other factors that may drive migrations, such as climate change, wars, pandemics, or natural disasters.

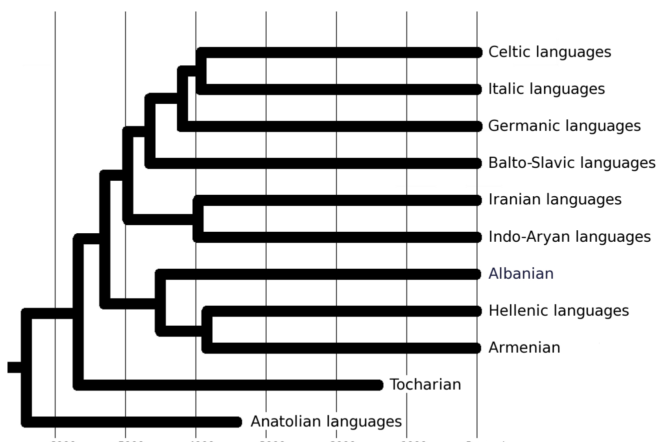

Figure 8: A phylogenetic tree of Indo-European languages according to Chang et al. (2015).

4.2 Anatolian Model

The results of the Anatolian simulation are provided in Figures 9 and 10, and with pre-evolution of Tocharian, in Figures 11 and 12. Note that the Anatolian scenario does not allow pre-evolving Hittite because its location would coincide with that of the rest of the PIE population. Consequently, Hittite is modeled as being closely related to geographically nearby languages such as Armenian. This makes the Anatolian model less accurate than the Steppe model, although the tree it discovers is similar overall to the Steppe tree, and many clades (Celtic, Balto-Slavic, approximate Indo-Iranian) are still recovered. We further observe that the Anatolian simulation converges very early resulting in very large cognate distances between sampled languages, which indicates that 7000 BC may be an unrealistically early date for the start of divergence.

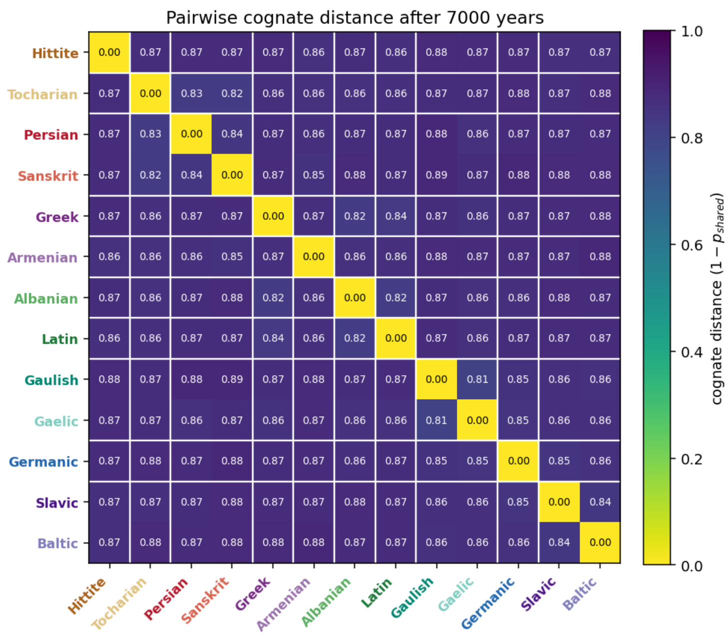

Figure 9: Heatmap of lexical distances at the end of the Anatolian simulation.

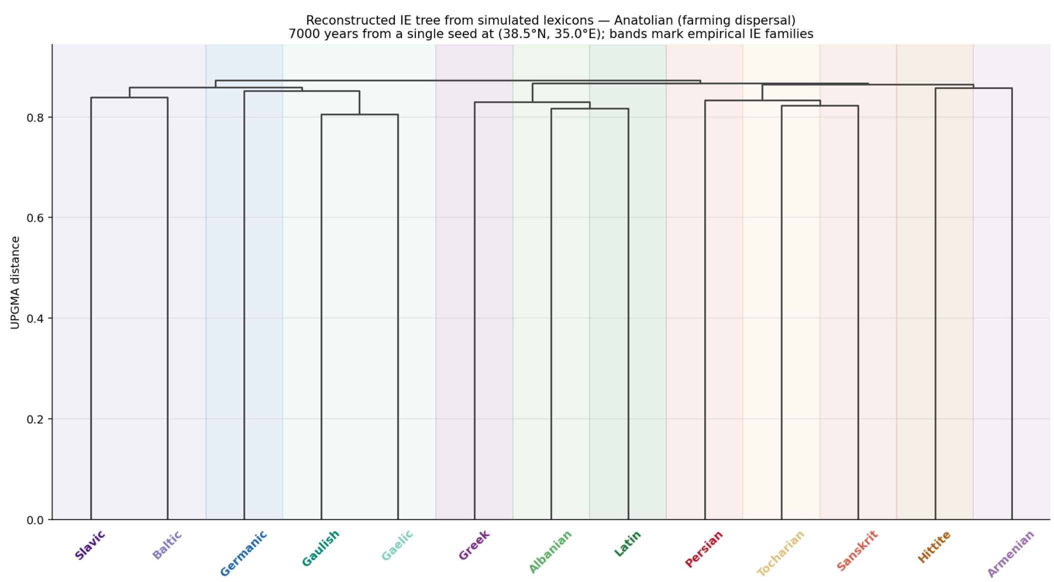

Figure 10: Dendrogram of Indo-European languages inferred from the heatmap in Figure 9.

5 Discussion

We have presented a computational simulation framework for modeling the spread and evolution of Indo-European languages. Upon comparing the simulations starting at the Steppe and Anatolian

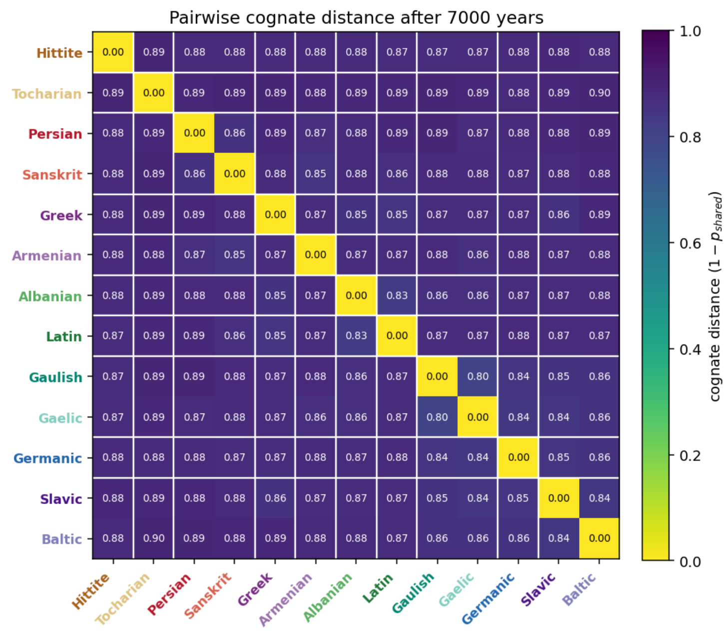

Figure 11: Heatmap of lexical distances at the end of the Anatolian simulation if Tocharian is pre-evolved.

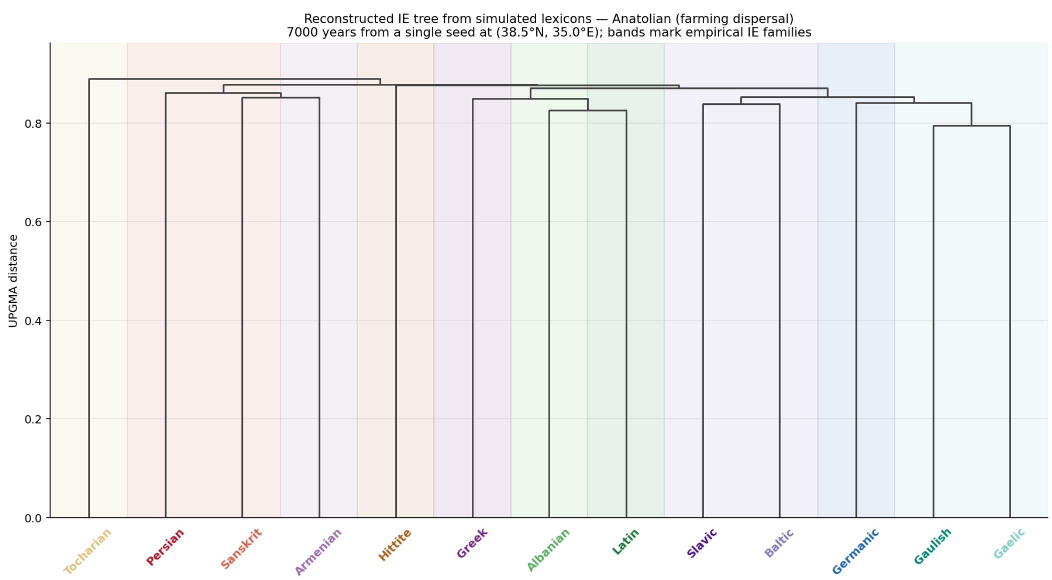

Figure 12: Dendrogram of Indo-European languages inferred from the heatmap in Figure 11.

homelands, we found that the Steppe model is more accurate, recovering the reference phylogenetic tree almost fully with only a few small errors. On the other hand, the Anatolian model has more significant issues, such as failure to recover Hittite as a basal outgroup and extremely early convergence. This result provides evidence in favor of a steppe origin of Indo-European languages.

A noteworthy limitation of our framework is its overreliance on geography-based, constant-rate population diffusion, together with its inability to model a wider variety of factors that can influence migrations. Future work could address this limitation by including factors like climate changes or disasters, and by varying the migration rate depending on location to model transitions between nomadic and settled lifestyles.

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